Shadow intersections in sporadic and tree geometries

Private coherent application; source-label obligations remain open

Abstract

One residue theorem and one complete witness search determine the geometric intersection properties of the supplied sporadic geometries. We consolidate the previously reviewed finite calculations, their actual geometry hypotheses and their exact family correspondences. Of the 57 numerical rows printed by Jacobs and Leemans, 53 agree with the current supplied-roster or retained-census calculations. The four differences concern two repeated rank-three reports: M_{12} and J_3. We also recover the Suzuki shadow split and the uniform tree criterion. Historical geometry suffixes remain separate, unresolved correspondences; the article's full labelled classification is not claimed recovered.

1 A single finite proof boundary

Our source is the complete sixteen-page edition of Jacobs and Leemans [JL], together with its twenty-file supplementary geometry archive. The common paper *Witnesses and orbits in finite incidence geometries* is a whole mathematical dependency. Its covering, orbit, double-coset, residue, sparse-coset, maximality and transporter arguments are written once there. The original orbit method is attributed to Jacobs and Leemans; the finite-group and block arguments are standard. Our common implementation and this assembly do not claim new priority for the source results.

For an enlarged Topic D, its entire accepted common paper *Stabilizers, extensions and the enumeration of finite geometries* remains a paid dependency. Its coset realization, panel recognition, local primitivity and augmentation theorems already contain part of the infrastructure used here. The additional shadow theory concerns geometric witness flags, normalized object–flag queries, sparse coset representatives and the rank-by-rank existential quantifier; these are not supplied by the old support-counting theorem. The detailed overlap ledger also marks repeated standard proofs as overlap. No existing common section is removed, and no overlap discount or compression factor is inferred from this assembly.

An input is an ordered family of actual subgroups in a specified acting group. The equivalence used for a finite roster is one automorphism of that group applied to the *whole* family, followed by a permutation of types. Equality of subgroup orders, intersection profiles, shadow answers, or aggregate counts never establishes this equivalence. The last active assignment in each archive file is parsed literally; it is kept distinct from the order of construction and from a historical geometry label.

Proposition 1. *Suppose a finite roster is given explicitly, and each input either passes the common geometry tests or is identified by an actual whole-family isomorphism with a previously accepted marked input having those properties. For every eligible equivalence-class representative, test all base residues of rank at least two and every remaining type by the common shadow theorem. The resulting all/exists quantifiers give exactly its* $(\mathrm{IP})_k$, $(\mathrm{WIP})_k$, *IP and WIP values. Summing these values over the exact family quotient gives the corresponding roster counts. A separate prior classification is needed to assert that the roster contains every geometry of the group.*

Proof. The eligible-extension equalities establish full flag transitivity, not merely transitivity on pairs. The common residue theorem then identifies the residue at each base flag with its

actual intersection-subgroup coset geometry, and transports it to every flag of the same type. In each residue the orbit/double-coset theorem exhausts all normalized element–flag queries and all eligible witness flags. Its sparse implementation changes only the representation of right cosets; empty flags still use the exact whole-universe cardinality. Thus each recorded type decision is the required equivalence, including negative decisions. Taking all types gives IP at that residue; taking at least one gives WIP, with the successful type allowed to vary between residues. Taking all residues of rank k, and then all $k \geq 2$, gives the stated properties. Panels are separately checked for geometry hypotheses; rank zero is never used for the existential WIP condition.

An actual family automorphism and type permutation induce an incidence isomorphism, which transports flags, shadows and witness flags. Therefore every tested property is constant on a family class. The exact transporter criterion makes the quotient neither omit a class nor count one twice. Summing its class values proves the roster counts, but gives no assertion about an input absent from the roster. This is precisely why the completeness premise is separate. □

The source population requires flag transitivity, firmness, residual connectedness, residual weak primitivity, local 2-transitivity and geometric IP_2. The first three and IP_2 are not inferred from an earlier subgroup-intersection check. Residual weak primitivity quantifies over every nonempty residue, including panels, and asks for at least one primitive type action of degree at least two. The exact double-coset maximality criterion permits nonfaithful actions. Local 2-transitivity checks the actual point-stabilizer orbits in each panel. Every excluded input below has a recorded failed hypothesis.

2 The combined sporadic calculation

Table 1 gives one row for each group and rank. Each triple lists IP, WIP without IP, and failure of WIP at the indicated residue rank. The last column takes the conjunction over all ranks, not just the top rank. Every retained class satisfies IP_2. The machine-readable table additionally retains all 57 literal source rows, their page and table locators, their original numbers, and the four precise differences.

Table 1: Exact current sporadic roster counts.

Group	Rank	N	$k=3$	$k=4$	$k=5$	$k=6$	Full
M11	3	13	6/1/6	–	–	–	6/1/6
M11	4	11	2/0/9	2/0/9	–	–	2/0/9
M12	3	17	7/1/9	–	–	–	7/1/9
M12	4	18	4/0/14	3/0/15	–	–	3/0/15
M12	5	1	1/0/0	1/0/0	0/0/1	–	0/0/1
M22	3	25	5/8/12	–	–	–	5/8/12
M22	4	16	5/4/7	3/1/12	–	–	3/1/12
M22	5	9	0/1/8	0/1/8	0/1/8	–	0/1/8
M23	3	13	5/1/7	–	–	–	5/1/7
M23	4	21	5/2/14	3/1/17	–	–	3/1/17
M23	5	10	2/1/7	1/1/8	1/0/9	–	1/0/9
J1	3	1	0/0/1	–	–	–	0/0/1
J1	4	2	1/0/1	1/0/1	–	–	1/0/1
J2	3	19	8/3/8	–	–	–	8/3/8
J2	4	3	1/0/2	1/0/2	–	–	1/0/2
J3	3	19	6/8/5	–	–	–	6/8/5
HS	3	30	8/5/17	–	–	–	8/5/17
HS	4	41	8/1/32	5/0/36	–	–	5/0/36
HS	5	8	1/0/7	1/0/7	0/1/7	–	0/1/7

Group	Rank	N	$k=3$	$k=4$	$k=5$	$k=6$	Full
HS	6	1	0/0/1	0/0/1	0/0/1	0/0/1	0/0/1

The entries for M_{12} rank three and J_3 are qualified below. Dashes denote a residue rank larger than the geometry rank. The $k = 6$ entry records the source's separate sentence about the unique rank-six HS geometry.

The total in this table is 278 representatives, compared with the printed population 277. This difference does not assert a new global sporadic classification. It consists of one fewer eligible M_{12} rank-three class and two additional inequivalent eligible J_3 inputs in the supplied archive. All other numerical source rows agree.

Mathieu inputs. The literal M_{11} archive has thirteen rank-three and eleven rank-four inputs. All twenty-four pass the source geometry hypotheses. Its final rank-three reorder is retained explicitly. For M_{12} there are twenty active rank-three, twenty rank-four and one rank-five input. Rank-three active entries 15,17,20 and rank-four entries 4,7 fail local 2-transitivity; the remaining seventeen, eighteen and one respectively give the table. All their residues and directional decisions were computed. The prior complete M_{12} subgroup census and the exact archive comparison are retained, not rerun; the one missing census class is treated below.

First and second Janko inputs. The three J_1 archive families match entire accepted marked families in the prior complete census: one rank-three and two rank-four cases. For J_2, twenty-five archive entries represent nineteen rank-three classes and only one of the three rank-four classes. The other two rank-four cases are supplied by the accepted complete Hall–Janko census, whose exact representatives are retained. Every archive entry, including duplicates, has a whole-family correspondence. No new subgroup census is performed here. Earlier accepted geometry predicates transfer only across those actual family maps; the shadow decisions are computed on all twenty-two class representatives.

The two larger Mathieu groups. Sixty-six active M_{22} families match fifty retained phase-seven classes, of ranks $3, 4, 5$ in numbers $25, 16, 9$. Six further construction entries lie outside those sixty-six active inputs; they are separately accounted for by their failed local 2-transitivity. Forty-four active M_{23} families match forty-four retained classes, of ranks $3, 4, 5$ in numbers $13, 21, 10$. The old complete census and residual weak-primitivity conclusions are retained; the maps identify the actual whole families, and all other current geometry and shadow checks are performed. Historical publication suffixes are not the phase-seven class indices.

Higman–Sims. All 124 literal active families—44 of rank three, 67 of rank four, 12 of rank five and one of rank six—pass actual source-filter checks. The exact full-automorphism/type quotient has $30, 41, 8, 1$ classes. Forty-four classes occur twice and thirty-six once. The full automorphism group has order 88 704 000. All intersection orders are necessary rejection invariants only; the actual subgroup transporters establish positive family identifications. Caching target stabilizer chains does not alter the permitted transporter group. The eighty representatives contribute 1,136 base residues including panels, 836 residues of rank at least two, and 2,204 directional shadow decisions. These are all used in the table. Completeness beyond the literal roster remains the classification cited in source reference [21].

3 The Mathieu exception does not come from a missing input

The retained complete M_{12} rank-three census has twenty-one classes; twenty have exact raw-archive correspondents. The remaining census class 18 is absent from the archive, but it cannot repair the source's eighteen-geometry locally 2-transitive population. This class number is our earlier census index, not a historical source name.

Proposition 2. *Let $G = M_{12}$ and let (H_1, H_2, H_3) be the marked subgroup family in retained census class 18. Its orders are*

$$|G| = 95040, \qquad (|H_1|, |H_2|, |H_3|) = (24, 660, 660), \qquad |H_1 \cap H_2 \cap H_3| = 6.$$

The geometry satisfies all of the above eligibility conditions except local 2-transitivity. It satisfies the full geometric intersection property. Its panel of type 1 is the action of A_5 on the ten two-element subsets of a five-element set. In particular, it cannot be added to the locally 2-transitive population.

Proof. The entire marked family is retained in `prior_representatives.g`. The unchanged exact residue procedure from the common shadow paper checks every base residue, including panels, and every remaining type. Flag transitivity makes these representatives exhaustive for residues of the same type. The resulting certificate verifies firmness, residual connectedness, residual weak primitivity, and the geometric intersection property in ranks two and three. These are exact finite group and coset calculations; no property is inferred solely from the displayed orders.

The failure of local 2-transitivity has a smaller independent proof. Put $B = H_1 \cap H_2 \cap H_3$ and $K = H_2 \cap H_3$. A bijective homomorphism $K \longrightarrow A_5$ maps B to

$$D = \langle (2\,4\,3), (1\,5)(2\,3) \rangle.$$

The program `identify_panel.g` checks this homomorphism and that D is exactly the stabilizer of $\{1, 5\}$. Thus the right B-cosets in K are identified equivariantly with the ten two-element subsets. This also gives a proof that is independent of the shadow procedure: `panels.g` constructs those ten right cosets directly and computes the action of their point stabilizer.

The stabilizer D has three orbits on the two-element subsets. The fixed subset is $\{1, 5\}$. There are three subsets disjoint from it, and six meeting it in one point. The displayed three-cycle permutes the complementary three points, while the displayed involution interchanges 1 and 5 and two of those three points. These generators act transitively on each of the latter two sets. The subdegrees are therefore $1, 3, 6$. Since the point stabilizer is not transitive on the other nine points, the panel action is not 2-transitive. The other two panels each have degree two. This proves the stated failure without relying on a previously recorded eligibility flag. □

Consequently, under the retained census and its exact family correspondences, the eligible rank-three population remains seventeen, with counts $7, 1, 9$. The source prints $7, 1, 10$ twice, in its rank-three and full-property tables. The omission and its non-2-transitivity were already recorded in the earlier census comparison; the complete shadow decision and the explicit small panel proof make the remaining obligation precise.

4 The third Janko roster has two additional classes

The final active assignment in `j3.rk3` swaps construction entries 8 and 9. All active indices in the following argument use that final assignment. The group has order 50 232 960. Every one of the twenty-one literal inputs passes actual geometry and local-filter tests on all 147 base residues, including panels. The full automorphism group has order 100 465 920 and the group has trivial centre. A verified noninner automorphism therefore completes the identity to a

transversal modulo inner automorphisms. The common simultaneous-conjugacy criterion tests both possibilities and every type permutation; it assumes no split extension. Saved positive maps retain the images of every ambient generator, the actual inner conjugator and the type permutation.

5 Explicit class separations and intersection results

For $J \subseteq \{1,2,3\}$ let $G_J = \bigcap_{j\in J} G_j$, with $G_\varnothing = G$. The multiset of all eight orders $|G_J|$ is invariant under group automorphisms and type permutations. On the supplied roster its buckets are the singleton entries and

$$\{7,8\}, \quad \{9,10,11\}, \quad \{13,14\}, \quad \{16,17\}.$$

There are 16 buckets. Actual complete-family transporters identify 13 with 14, and 16 with 17; their type permutations, automorphism images of every ambient generator, and the conjugating permutations are retained as exact data. The other repeated buckets split, with the following independent witnesses.

Entries 7 and 8 each have a unique parabolic of order 96; these parabolics contain respectively 27 and 15 involutions. Entry 9 has two parabolics of order 96, each with 27 involutions, whereas entries 10 and 11 have two such parabolics each with 15. These actual counts are obtained from complete conjugacy-class rosters in the small parabolic groups; group isomorphisms preserve them. Finally entry 10 satisfies full IP, while entry 11 fails WIP. The complete shadow certificates, with the failed directional intersections, therefore separate this last pair as well. These tests independently support the negative transporter decisions and give $16 + 1 + 2 = 19$ distinct supplied classes without relying on historical names.

Proposition 3. *Modulo entire-family automorphisms of the retained J_3 action and type permutations, the supplied roster has* 19 *eligible classes. Every class satisfies* IP_2*. Full IP, equivalently* IP_3 *here, holds in six classes; eight more satisfy WIP; five fail WIP.*

Proof. The order profiles separate different buckets. The explicit transporters identify exactly the two stated pairs, and the preceding invariants separate the other entries, so the quotient has 19 classes. All supplied entries were checked for the source geometry hypotheses, so these predicates transfer across the actual family isomorphisms.

For each representative, the accepted residue theorem enumerates all seven proper fixed-type subsets. Flag transitivity transports each base residue to all flags of those types. Panels are checked for geometry properties and only ranks two and three enter IP and WIP. This gives 133 base residues including panels, 76 of rank at least two, and 171 directional shadow decisions. Their actual all/exists quantifiers are reconstructed from the recorded values. The positive active representatives are

$$2, 3, 4, 7, 8, 10;$$

the weak-only representatives are

$$1, 5, 6, 12, 16, 18, 19, 21;$$

and the nonweak representatives are

$$9, 11, 13, 15, 20.$$

All rank-two residues satisfy IP, so the rank-three decision and full residual decision agree. This proves the asserted supplied-roster counts. □

Thus the supplied-roster split is $6, 8, 5$, whereas the source prints $5, 8, 4$. The global earlier classification in source reference [19] is retained separately; no unprovided input or historical suffix is identified by the number nineteen.

6 Historical labels and the remaining source obligation

Tables 2,4,6,8 name particular earlier geometries, not merely totals. The present source map therefore preserves 232 positive-label claim occurrences as explicit correspondence obligations. The label fields in current computed classes remain null. A matching property profile is insufficient evidence: many inequivalent geometries share it.

The archive contains only two commented M11 placeholder assignments; the other eighteen geometry files have no label comments. The M11 placeholders do not account for every printed suffix, including 3.18. The README gives one explicit example: HS rank-four active position 2 is said to represent 4.1. It cites reference [20], whereas the article uses [21] for HS. More seriously, the literal active position 2 is construction position 2, global HS input 46, exact family class 32. Its current certificate fails WIP. The article lists 4.1 as satisfying IP. These statements cannot jointly identify the literal current family with that printed positive result. We preserve the documentary contradiction and do not silently repair a label by permuting rows.

The appropriate conclusion is limited: all supplied inputs have been processed, the stated corrected roster counts and witnesses are proved, and the full historical naming correspondence remains open. The complete article is therefore not yet assigned recovered-original credit by this assembly. No separate source-label task is assumed completed elsewhere.

7 The Suzuki application

The following local enumeration uses the whole thin-triple and residue theory in the common paper. It retains the specified degree-65 Suzuki model and the earlier classification assertion that the relevant rank-three RWPRI geometries are thin; it does not claim recovery of the unavailable earlier classification article.

A finite enumeration with an explicit orbit reduction

The ambient permutation group is the previously validated degree-65 model used in the rank-two Suzuki calculation. We retain that model and GAP's exact finite group algorithms. The computation checks order 29120 and simplicity, one conjugacy class containing all 455 involutions, and an automorphism group of order 87360. Its action on those involutions has trivial kernel. It also computes representatives of all maximal subgroups, whose orders are $448, 52, 20, 14$. These are explicit computational premises; their implementations have not been formalized in Lean here.

Let $\mathcal{A}$ be the computed automorphism group acting on the 455 involutions, numbered in a fixed order. Fix the first involution. For its stabilizer $\mathcal{A}_1$, take one representative b from each orbit on the other involutions. For each b, take one c from each orbit of $\mathcal{A}_{1,b}$ on the remaining involutions. This gives all ordered triples up to $\mathcal{A}$, exactly once.

To forget the type order, normalize each of the six permutations of a triple by the same stabilizer procedure and retain the smallest resulting pair of indices (b, c). The normalization is independent of transporter choices: after placing the first entry at 1, any two choices differ by $\mathcal{A}_1$; after placing the second entry at b, they differ by $\mathcal{A}_{1,b}$. Thus the final orbit of the third entry is fixed. Equality of the minimum pairs is equivalent to equivalence under $\mathcal{A}$ and a permutation of the three types.

For each retained triple the program checks generation, the exact subgroup intersections, all three product equalities (2), and the primitivity conditions of Lemma 13. Top primitivity is checked by conjugacy to the computed maximal subgroups, not inferred from a subgroup order. The resulting counts are

Stage	Count
Ordered triples modulo $\mathcal{A}$	1083
After forgetting the three type names	195
Generating triples	186
Triples satisfying all marked subgroup intersections	186
Flag-transitive thin geometries	183
Residually weakly primitive geometries	147

The three rejected generating intersection systems show why a subgroup intersection test alone does not establish the required geometry.

For these thin connected geometries the type-preserving automorphism group is exactly G. Indeed an automorphism fixing one chamber fixes its unique neighbors across every panel, and then every chamber by connectedness. The full type-preserving group is consequently semiregular on chambers, while G is already transitive. It can have no more elements than G. Thus an isomorphism between two such geometries induces an automorphism of G and a permutation of types, exactly the equivalence used above.

Shadow tests and their limits

For each retained triple, the common shadow theorem expresses every shadow as an orbit of a chamber stabilizer intersection in $H_i\backslash G$. It checks all normalized object–flag pairs and every allowable witness flag. The implementation is the earlier reviewed shadow implementation, with the ambient order and input triples changed. It caches conjugate coset actions: if $H' = H^g$, the base point changes from p to p^g. The action is unchanged, and the stabilizer of the new point is checked to equal H'. No shadow or witness is removed by this optimization.

The full eligible-extension test is repeated in all three coset actions for each geometry, providing 1764 flag-extension identities in addition to the product tests above. Each of the 441 type tests is recorded. The resulting numbers are 27 with all three types passing, 48 with some but not all types passing, and 72 with no type passing. All rank-two residues are polygons of lengths $2m_i$ with $m_i \in \{2, 5, 7, 13\}$. For $m_i > 2$ there is no quadrilateral: shadows of distinct opposite-type vertices meet in at most one object, which itself is a witness whenever present. If $m_i = 2$, the residue is the complete bipartite graph $K_{2,2}$; the two opposite-type shadows are the entire target type, with the empty flag as witness. Flags already containing the target type have singleton shadows and use that object as witness; the empty input flag has the whole type as shadow. This proves the rank-two intersection property in both cases. Rank-one residues are automatic. Consequently the rank-three answers are also the full residual answers.

The orbit and product arguments enumerate all *thin* geometries with the stated hypotheses. They do not prove that every residually weakly primitive rank-three geometry of Sz(8) is thin. That separate classification assertion is retained from the earlier classification cited by Jacobs and Leemans. We claim no recovery of its unavailable manuscript. The present result is the reported shadow split on its 147 thin inputs.

The five eligible square-residue cases are included, not discarded as degenerate polygons. Among all 183 thin flag-transitive cases there are seven with a square residue: five pass RWPRI and two fail it. The seven is compatible with the previously accepted Topic D regular Suzuki polyhedron count $P(3) = 7$ up to duality. That older count and its proof remain attributed to that earlier work; they are a comparison, not a new premise or new result obtained from matching this number.

8 Transfer to inductively minimal geometries

We use the precise convention recalled in [BC98]: a pair consisting of a finite firm rank-r geometry with connected diagram and a flag-transitive group G is called *minimal* when $|G| \leq (r+1)!$. The word refers to this numerical threshold, not to minimality in a subgroup ordering. Inductive minimality requires that for each connected type subgraph J and every flag F of complementary type, the residue pair (Γ_F, G_F) is minimal in this sense. We retain the foundational classification of Buekenhout, Cara and Dehon, in the precise form recalled by Buekenhout and Cara [BC98]: an inductively minimal pair of positive rank r has group $\mathrm{Sym}(r+1)$; its connected diagram has no induced cycle longer than three, each edge belongs to a unique maximal clique, and each vertex belongs to one or two maximal cliques. Every such connected diagram determines one pair up to isomorphism. The one-vertex diagram is treated separately. The proof of this foundational classification is not replaced here.

These diagrams are precisely line graphs of trees. To see the required direction, form a vertex for each maximal clique. A diagram vertex in two cliques gives an edge between their clique vertices; a vertex in one clique gives an edge to a new leaf. Distinct cliques meet in at most one vertex, because a common edge would violate uniqueness of its clique. The constructed graph is simple and connected, and its line graph is the original diagram. A shortest cycle of length at least four would give an induced cycle on its edge labels in the diagram. A triangle would place three pairwise adjacent labels in one clique, contradicting unique ownership of the pairwise edges by the three cliques already represented at the cycle vertices. The constructed graph is therefore a tree. Its reconstruction is inverse to taking the nontrivial vertex stars of a tree as line-graph cliques. The one-vertex diagram corresponds to the one-edge tree.

For the geometry in Lemma 8, two remaining edges give either a six-cycle residue or $K_{2,2}$, according as they meet or are disjoint. Its diagram is consequently the line graph of T. Every residue with connected diagram and m remaining types comes from a single subtree on $m+1$ vertices, and its group is $\mathrm{Sym}(m+1)$. Its order is exactly $(m+1)!$, satisfying the stated source definition of minimality. Thus $\Gamma(T)$ is inductively minimal. The retained uniqueness identifies every inductively minimal pair with one of these tree geometries.

Corollary 4. *An inductively minimal geometry has the intersection property exactly when its diagram is linear. In particular this holds for the* $\mathrm{Sym}(n)$ *geometries of rank* $n-1$ *with* $2 \leq n \leq 7$.

Proof. Apply the retained classification, the construction just proved, and Theorem 11. The line graph of a tree is a path exactly when the tree itself is a path: a vertex of degree at least three would give a triangle in the line graph. □

9 Exact executable, retained and formal boundaries

The complete finite programs and actual input data are mathematical dependencies for the computational conclusions. Their role inventory distinguishes source parsing, actual geometry eligibility, family transport, shadow decisions and result aggregation from optional validation and operational rendering. Successful log summaries alone are not substituted for the full executable files. The old accepted subgroup-census programs, models, completeness arguments and scoped review records remain dependencies wherever the preceding applications retain them; those computations are not silently free prior results. GAP's exact finite-group algorithms and installed catalogue inputs are trusted computational infrastructure, with their pinned versions and limits retained.

No unchanged large census or completed full eligibility batch needs to be rerun to assemble this manuscript. The current aggregation recomputes every all/exists quantifier and all 57 source comparisons directly from the accepted directional records. For J_3, the optional five-program route recomputes family transport, saved-map data, small local invariants and shadows; it retains

the four actual eligibility batches. A stale opening docstring in the frozen runner says maximality is rerun; its README and mathematical report give the correct boundary, retained here.

The source prints concrete Magma programs in Section 6 and an executable algorithm in Figure 1. Future manuscript-only lengths must apply the same exclusion rule to these listings and to ours, while retaining mathematical explanation on both sides. Whole required software and data remain visible separately; missing source software costs are unknown, never zero. Table 9 and the Suzuki twelve-second statement are historical execution reports, not new mathematical or performance claims. This assembly supplies no length ratio.

The accepted four-lemma covering formal module certifies its exact set/flag covering equivalence. The inherited Boolean-shadow module proves the interval description and at-most-two-element witness for finite query families. Neither proves the whole orbit algorithm, sporadic computations, thin Suzuki census, tree identification or the retained classifications in Lean. Their complete files and exact declaration scope remain separately bound. There is no new conjecture credit: the all-degree tree result was already attributed by the source to Cara and Leemans.

References

[JL] P. Jacobs and D. Leemans, An algorithmic analysis of the intersection property, *LMS J. Comput. Math.* 7 (2004), 284–299.

[BC98] F. Buekenhout and P. Cara, Some properties of inductively minimal geometries, *Bull. Belg. Math. Soc. Simon Stevin* 5 (1998), 213–219. The foundational 1997 classification is retained in the precise form recalled there.

[CL05] P. Cara and D. Leemans, On inductively minimal geometries that satisfy the intersection property, *J. Combin. Theory A* 111 (2005), 327–330. The source attributes the uniform conclusion to this prior work; the unavailable full article receives no original credit here.